Land Cleared by Humanitarian Demining in Colombia Before and After the 2016 Peace Agreement

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Overview

In this project I tested whether humanitarian demining in Colombia cleared more land per month after the 2016 peace agreement than before it, using the government's open record of demining results, *Situación de Desminado Humanitario en Colombia* [Humanitarian Demining Situation in Colombia] (Presidencia de la República de Colombia, 2024), available at https://www.datos.gov.co/Inclusi-n-Social-y-Reconciliaci-n/Situaci-n-de-Desminado-Humanitario-en-Colombia/jbqm-s5tn. The Final Agreement between the Government of Colombia and the FARC-EP was signed on November 24, 2016 (United Nations Security Council, 2017), which divides the 14 years I tested, 2010 to 2023, into seven years before and seven after. The question matters to me because, from 25 years in humanitarian mine action and arms control, I believe that clearance can go on during a war but is far more effective in peace: cleared land is not mined again behind the teams, deminers work with more safety and security, and security forces no longer needed for the fighting become a source of personnel for peace support work such as clearance. The notebook, data and environment file are in a public GitHub repository, https://github.com/travisrcarlson/statistical-analysis-project.

Dataset Description

The dataset is Colombia's official record of the results of humanitarian demining, published by the Presidencia de la República de Colombia on the national open data portal under the CC BY-SA 4.0 licence; I used a snapshot downloaded on October 18, 2026, whose rows the publisher last updated on January 19, 2024. It has 5,721 rows and 18 columns, covering 2004 to 2023 and 435 municipalities in 31 departments and the capital district, Bogotá. Each row holds,

for one municipality, month, demining type and operator, either results (land cleared in square metres, explosive items found, surveys and clearance operations) or counts of hazardous areas; 16 rows hold neither. The key variables were land cleared, which I added up into national monthly totals, and the year and month, which place each record before or after the agreement; the operator, demining type and department gave context. No cell is empty and no row repeats another, but 1,183 combinations of municipality, month, demining type and operator appear twice; the two rows never report the same measure, apparently because the file joins two kinds of record, so I added them up rather than deleting them as duplicates.

Methods

Descriptive Statistics

I followed the initial data analysis approach that Lusa et al. (2024) describe: the research question and the analysis plan come first, and the data checks then examine the data's properties for that plan while refraining from analyses related to the question, because decisions made after looking at the data can bias the results. I therefore fixed the question, the unit of analysis, the test and a check on yearly totals before running the test, and kept the descriptive statistics to properties of the data rather than comparisons between the periods. The question itself was chosen after I had seen the yearly totals, so the plan guards against choices made during the analysis, not against the choice of question (see Limitations). Summary statistics showed that land cleared is zero in 72% of records and heavily right-skewed, so I described every measure with its share of zeros, its median and high percentiles, not only its mean. For the national monthly totals, I screened what the test depends on: months with no record, which the plan counts as zero; months with zero land cleared; the skewness of each period's distribution; and

how strongly each month resembles the month before (lag-1 autocorrelation), since Lusa et al. (2024) recommend evaluating how strongly repeated measurements are correlated over time.

Visual Models

I chose three charts, each answering a different part of the question. A line chart of land cleared per year (Figure 1) shows when any change happened and puts the start-up years in context. A box plot of the monthly totals by period (Figure 2) compares the two groups of months that the test compares, with their medians and middle halves, and shows every month as a dot. Histograms of the monthly totals by period (Figure 3) show the shape of each distribution, which illustrates why the plan chose a rank-based test. Each chart trades something away: the line chart hides the month-to-month variation that the test works with, the box plot loses the order in time, and the histograms make the centres harder to compare (notebook Section 4.4).

Hypothesis Test

The null hypothesis (H_0) was that the national monthly totals of land cleared in 2010–2016 and in 2017–2023 come from the same distribution; the alternative (H_1) was that they differ, tested two-sided at $\alpha = .05$, with 84 months in each period. I left out 2004–2009, when operations were starting, with 1 to 50 records a year. I used the Mann–Whitney *U* test because it works on ranks, so a few very large months cannot dominate it, and because it matches the question, which asks whether months after 2016 tended to clear more land; the monthly totals are also bounded at zero and mildly right-skewed. The test's parameter, which Fay and Proschan (2010) call the Mann–Whitney functional, is the probability that a randomly chosen month from one period exceeds one from the other, counting ties as half. I report that probability and the rank-biserial correlation, which rescales it from −1 to 1, alongside *U* and the *p*-value, because a

p-value indicates how incompatible the data are with the null hypothesis but does not measure the size of an effect or its importance (Wasserstein & Lazar, 2016).

Assumptions

The test assumes independent observations, and failure of that assumption can have a large effect on a test's validity (Fay & Proschan, 2010). Consecutive months were correlated (lag-1 autocorrelation of .29 before the agreement and .45 after), so the monthly *p*-value is probably too small; as planned, I repeated the test on yearly totals, seven years against seven, which depend far less on each other. The test is valid for the null hypothesis that both periods share one distribution, but the periods differ in spread, and a difference in spread can also make the test significant; when spreads differ, the Mann–Whitney test is not a valid test of whether one period's values tend to be larger, while the Brunner–Munzel test is valid in large samples (Fay & Proschan, 2010). I therefore added a Brunner–Munzel test as a check that was not in the plan. Finally, three months in the test window have no record and count as zero, as the plan specified. December 2023 probably had not been reported when the data were updated, and counting it as zero can only make a rise harder to find. January 2011 and January 2012 count against the earlier period and so slightly favour a rise, although seven of the eight months with no land cleared before 2017 fall in January or February, so they may be real pauses. I also repeated the test without these three months, as a second check that was not in the plan.

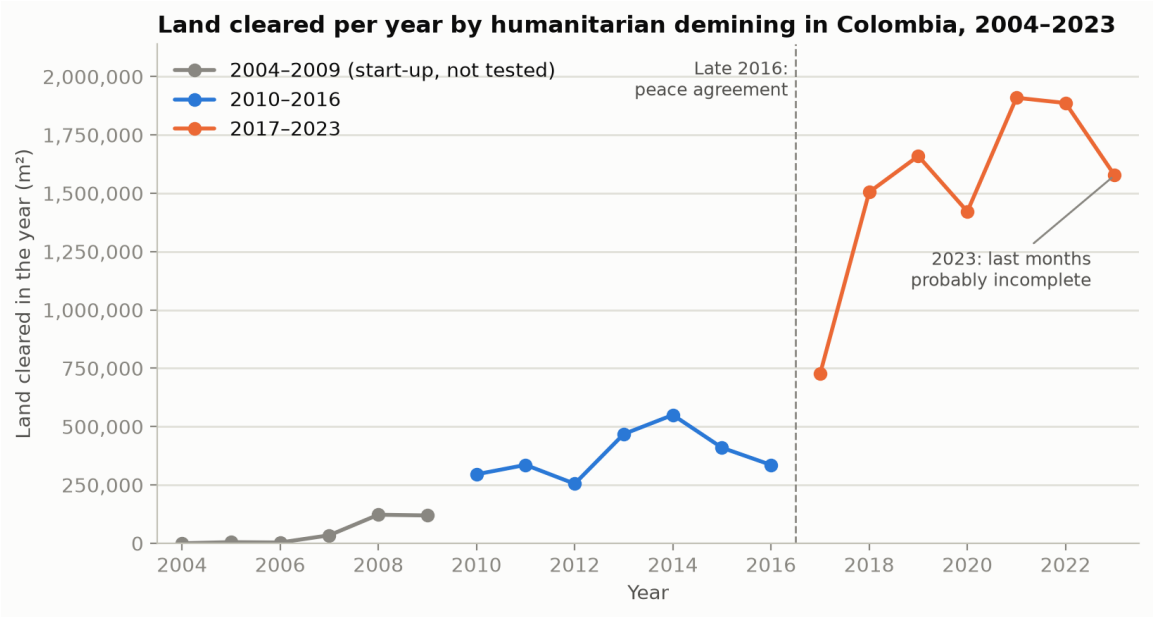
Results

Land cleared stayed below 125,000 m² a year until 2009, ranged from 255,010 to 549,931 m² a year in 2010–2016, then rose to 726,267 m² in 2017 and to between 1,421,677 and 1,909,198 m² a year in 2018–2023 (Figure 1). Every year after 2016 cleared more land than any

year before. The 2023 total of 1,576,857 m² is probably an undercount, because records thin out from September 2023.

Figure 1

Land Cleared per Year by Humanitarian Demining in Colombia, 2004–2023

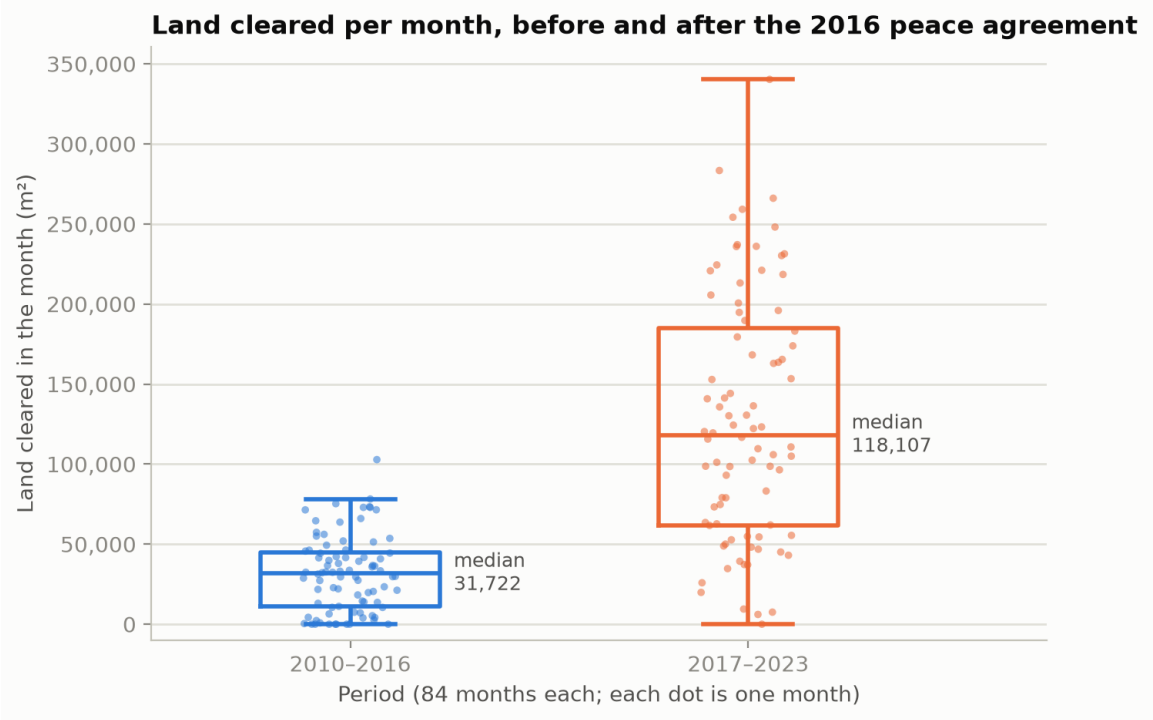


Note. From notebook Section 4.1. The years 2004–2009, when operations were starting, are shown for context and left out of the test. Data from Presidencia de la República de Colombia (2024).

The monthly totals show the same change (Figure 2). The median month rose from 31,722 m² in 2010–2016 to 118,107 m² in 2017–2023, about 3.7 times as much, and the middle half of months moved from 10,979–44,727 m² to 61,895–184,724 m², with no overlap between the two ranges. Months after 2016 also varied much more, from 0 to 340,263 m², and eight of the nine months with no land cleared fall before 2017.

Figure 2

Land Cleared per Month Before and After the 2016 Peace Agreement

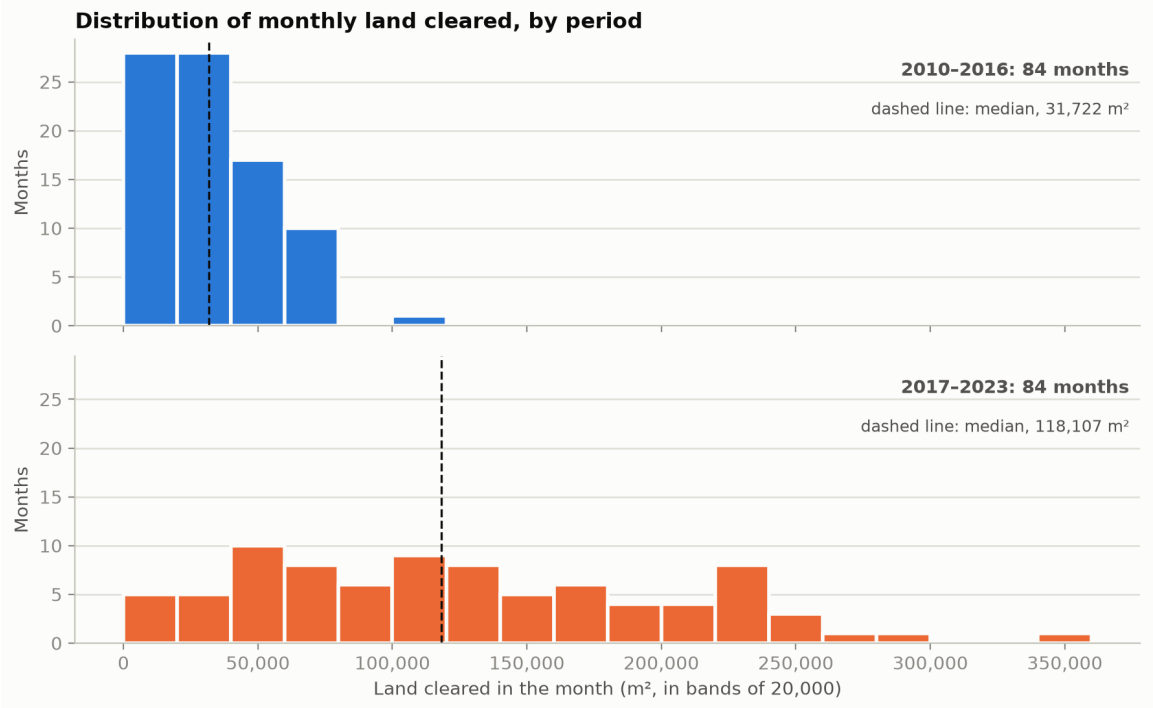


Note. From notebook Section 4.2. Each dot is one month (84 in each period). Boxes show the middle half of the months, with the median as the line inside; whiskers reach the furthest month within 1.5 times the box height of the box. Data from Presidencia de la República de Colombia (2024).

The distributions differ in shape as well as in level (Figure 3). Before 2017, 73 of 84 months (87%) cleared less than 60,000 m²; after 2016 the months spread across the whole range, and only 20 of 84 (24%) fell below 60,000 m². Of the three charts, the box plot best supports the question, because it shows exactly what the test compares.

Figure 3

Distribution of Monthly Land Cleared by Period



Note. From notebook Section 4.3. Months are counted in bands of 20,000 m²; the dashed line marks each period's median. Data from Presidencia de la República de Colombia (2024).

The Mann–Whitney test rejected the null hypothesis, $U = 6{,}314$, $p < .001$. The Mann–Whitney probability was .895, with ties counted as half: in about 89% of the 7,056 possible pairs of one month before and one month after the agreement, the month after cleared more land (rank-biserial correlation = .79). The yearly check agreed: every year in 2017–2023 cleared more land than every year in 2010–2016, $U = 49$ of 49 pairs, exact $p = .0006$. Because months are correlated, the monthly p-value probably overstates the evidence, but the yearly check does not depend on treating months as independent and reaches the same conclusion. The two checks added after the plan agree as well: the Brunner–Munzel test gave $p < .001$, and without the three months that have no record the Mann–Whitney test gave $U = 6{,}144$ of 6,806 pairs, $p < .001$ (notebook Section 5.5).

These results show that more land was cleared per month after the agreement; they do not show why. Other things changed at the same time. The number of operators doubled after 2016, but most of the extra land came from operators already at work before 2017: BRDEH, the Army's humanitarian demining unit, accounts for 74.7% of the increase and The HALO Trust for 14.2%, while the seven operators that cleared no land before 2017 cleared 6.5% of the land in 2017–2023 (notebook Section 5.6). Staffing, funding, access and reporting may also have changed, and this dataset does not measure them.

Interpretation for a Non-Technical Audience

In plain terms, after the peace agreement the demining teams cleared nearly four times as much land in a typical month as before: about 118,000 square metres a month, roughly the area of 16 full-size football pitches of 105 by 68 metres, against about 32,000 square metres, roughly four pitches. This was not a matter of a few unusual months: every year after 2016 cleared more land than any year before, and a difference this large would be very unlikely if nothing had changed between the two periods. Uncleared land stays dangerous to work on for generations, as other countries show: more than a century after the First World War, Belgium's army bomb-disposal unit still removes over 150 tonnes of unexploded ordnance a year, and farmers still hit shells with their machinery (Coe, 2023). Clearing land frees it for farming and other economic uses; in my experience that helps to stabilise communities, because when economic opportunity returns to the local population, people start to thrive. The data show that clearance rose after the agreement, but not why. One explanation, from my experience, is that clearance is far more effective in peace: cleared land is not mined again behind the teams, deminers can work in more safety, and security forces no longer needed for the fighting can take on peace support work such

as clearance. That the Army's demining unit cleared most of the extra land fits this explanation, but testing it would need data on staffing, funding and security that this dataset does not hold.

Limitations and Potential Bias

Limitations of the Data

- **Land cleared measures output, not risk.** A square metre of easy farmland counts the same as a square metre of rocky ground or jungle, and the data do not say how dangerous the land was or how hard it was to clear.
- **Cleared is not released.** Land release is the process of identifying, defining and removing all presence and suspicion of explosive ordnance through non-technical survey, technical survey and/or clearance, and survey should normally take priority over clearance (United Nations Mine Action Service [UNMAS], 2019). This dataset records the land cleared only; its survey columns count surveys, not the land they release. Clearance also has layers: it is done to a specified depth or other parameters set by the national authority (UNMAS, 2019), and in my experience land meant for farming has to be cleared deeper than the surface, so one square metre of clearance does not always equal another.
- **Mines or unexploded ordnance.** Items found are split into antipersonnel mines, unexploded ordnance and improvised explosive devices, but land cleared is a single figure, so the data cannot say whether land was cleared of mines or of unexploded ordnance, which are different tasks.
- **Missing or late reports.** Records thin out at the end of 2023, and under-reporting in remote areas cannot be measured from this file, so some months and places may look smaller than they were.

- **Limits of the design.** Months are correlated, which the yearly check addresses. I chose the question after seeing the yearly totals, so the test confirms a visible pattern rather than a blind prediction. Other things changed after 2016, including the number of operators, so the rise cannot be attributed to the agreement alone.

Potential Bias and Misuse

- **Chasing square metres.** If operators are judged by the square metres they clear, they are pushed toward easy land. In my experience the easiest land is usually the farmland, while the complex clearance lies in the rocky ground next to it, on paths through rural areas and in dense jungle, so a target in square metres would leave those harder areas, including the paths people use, for last.
- **Funding by headline numbers.** Donors who compare operators by square metres can penalise teams working in harder terrain or denser contamination. The same risk applies between countries: mine action funding is becoming concentrated in a small number of countries, with 72% of international assistance going to the 10 largest recipients in 2024 and a significant share going to Ukraine since 2022, while support to other States Parties to the Mine Ban Treaty has declined (International Campaign to Ban Landmines – Cluster Munition Coalition, 2026). Communities with smaller numbers may be in greater need.
- **Changes in reporting.** Records rose from 92 in 2010 to 956 in 2018. Part of that rise reflects more work, but if reporting also became more complete after 2016, the rise in land cleared is overstated, and this file cannot separate the two.
- **Overclaiming the cause.** The rise fits the agreement in time, but using it to argue that the agreement worked, or failed, goes beyond what this test shows.

- **False sense of safety.** Totals and maps of cleared land can lead people to assume that nearby land is safe too, when clearance covers only the area, and the depth, that was cleared (UNMAS, 2019).

Next Steps

Three analyses would take this further. A time-series model of the monthly totals that allows for correlated months could show when the change began and how large it was. Adding the land released through non-technical and technical survey, from other sources, would measure land release as a whole rather than clearance alone. Comparing Colombia with other countries would put the result in context, provided the comparison does not become a league table that pulls funding away from communities with smaller numbers but greater need.

Citation Requirements

I used citations for statistical concepts, the choice of test and its assumptions, and the limitations and risks, but not for my own results, which come from the notebook. The required article, Lusa et al. (2024), supports the analysis plan and the data screening. Fay and Proschan (2010) support the test's effect size, the discussion of its assumptions and the added Brunner–Munzel check, and Wasserstein and Lazar (2016) support how I report *p*-values. UNMAS (2019) supports the points on land release and clearance; the dataset, the peace agreement, the Belgian example and the funding figures are cited to their sources. The references follow APA 7.

I used Claude (Anthropic). I ran the notebook and checked its results.

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